

RAHAL NANAYAKKARA

PhD student at the Cyber-Physical Systems Lab, University of California, Los Angeles

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EDUCATION

University of California, Los Angeles (UCLA)

PhD in Electrical and Computer Engineering

MS in Electrical and Computer Engineering

Advisor : Prof. Paulo Tabuada

Sep 2023 - Present

Sep 2023 - Jun 2025

GPA: 4.00/4.00

University of Peradeniya (UoP) - Sri Lanka

BSc. in Electrical and Electronic Engineering (Top of Class)

2018 - 2023

GPA: 4.00/4.00

Selected Classes (UCLA) : Linear Dynamic Systems, Nonlinear Dynamic Systems, Optimal Control, Geometric Nonlinear Control, Convex Optimization, Decision Making in Stochastic Systems

EXPERIENCE

University of California, Los Angeles - Graduate Student Researcher

Sep 2023 - Present

Worked on several projects in areas of estimation, nonlinear systems and safety-critical control.

- *Robust Safety Critical Control* - Currently working on developing algorithms for synthesizing safety critical controllers in the presence of state uncertainty using modified control barrier functions. (In collaboration with AMBER Lab at Caltech, Pasadena)
- *State Estimation* - Worked on a computationally light, nonlinear state estimation algorithm that provides online error bounds from sampled output data. [[repo](#)]
- *Other Projects* - Worked on a previously developed LiDAR localization algorithm with provable error bounds. Currently also working on using control theory to model the spread of epidemics on networks. Implemented Model Predictive Controller for autonomous cars for waypoint tracking in simulation environment.

University of Peradeniya - Research Assistant

Mar 2023 - Aug 2023

- Worked on a biomedical engineering project for fetal health monitoring based on sensory data from abdomen mounted inertial sensors.

Vega Innovations - Electronic Engineering Intern

Aug 2022 - Nov 2022

- Worked on flux weakening control for high speed operation of Permanent Magnet Synchronous Motors in electric vehicles.
- Implemented over the air firmware updates on embedded microcontrollers.

Sri Lanka Institute of Nanotechnology - Electrical Engineering Intern

Nov 2019 - Feb 2020

- Designed electrical power distribution system for a graphene oxide production plant.
- Production and assembly of enhanced lead acid battery prototypes.

SKILLS

Programming Languages

Python, MATLAB, Java, C/C++

Software Tools

Git, Numpy, PyTorch, ROS, OpenCV

Hardware Skills

PCB design & development, wiring harness construction, SMD soldering

Technical Knowledge

Linear and Nonlinear Control Theory, Optimization, Machine Learning

SELECTED AWARDS

1. **C.A. Hewavitharana prize for the best performance in engineering, 2023:** Awarded for overall best academic performance in the school of engineering, University of Peradeniya.
2. **CEB Gold medal for best performance in electrical & electronic engineering, 2023:** Awarded to top of the class in the Department of Electrical and Electronic Engineering.
3. **Prof J.A. Gunawardena prize for the best performance in control engineering, 2023:** Awarded for best academic performance in controls related subjects (s-domain control, discrete and continuous time state space based control, basic nonlinear control).
4. **Bartholomeusz Prize for Engineering Mathematics, 2020 & 2021:** Awarded for best performance in mathematics modules in the 1st and 2nd year of the undergraduate program.
5. **Bronze Medal at the Sri Lanka Mathematics Olympiad, 2017:** Ranked 5th nationally, with High Distinction. Conducted by the Sri Lanka Olympiad Mathematics Foundation.

SELECTED ACHIEVEMENTS

IEEEExtreme 14.0 - 24 hour global algorithmic programming competition National Rank - 2, World Rank - 68	2020
IEEEExtreme 13.0 - 24 hour global algorithmic programming competition National Rank - 5, World Rank - 130	2019
MoraXtreme 4.0 - 12 hour algorithmic programming competition Placed 1st out of 150 teams in Sri Lanka	2019
ACES Hackathon - Intra university hackathon 1st place in Travel and Safety Category Project : Neural Network based CCTV System for tracking individuals and unattended baggage	2019
SLIIT Robofest - Autonomous maze navigation (Micromouse) robotics competition Placed 3rd out of 50 teams	2019
G.C.E.(A/L) - Competitive national examination for university entrance District Rank - 1 out of 2,987. National Rank - 14 out of 33,608. Z - score - 2.7522	2016

PUBLICATIONS

Conference Papers

1. J.P. Silvestre, **R. Nanayakkara** and P. Tabuada, "Nonlinear Observers with Tighter Online Error Bounds," 2024 *IEEE 63rd Conference on Decision and Control (CDC)*.
2. L.B.I.P. Thilakasiri, D.M.P.M. Alwis, **R.T. Nanayakkara**, G.M.R.I. Godaliyadda, M.P.B. Ekanayake and J. V. Wijayakulasooriya, "Fetal Movement Identification Using Spectrograms with Attention Aided Models and Identifying a Set of Correlating Parameters with Gestational Age," 2023 *IEEE 17th International Conference on Industrial and Information Systems (ICIIS)*.
3. L.B.I.P. Thilakasiri, D.M.P.M. Alwis, **R.T. Nanayakkara**, G.M.R.I. Godaliyadda, M.P.B. Ekanayake and H.M.V.R. Herath, "Integrated Video Based Crowdedness Forecasting Framework with a Review of Crowd Counting Models," 2021 *IEEE 16th International Conference on Industrial and Information Systems (ICIIS)*.

Workshop Presentations

1. **Rahal Nanayakkara**, Ersin Das, Xiao Tan, Aaron Ames and Paulo Tabuada. "Safety Under State Uncertainty: Robustifying Control Barrier Functions," 45th *Southern California Control Workshop 2025*.

SELECTED PROJECTS

LiDAR localization

UCLA, 2025

Performed comparison of LiDAR localization algorithms based on experimental data collected in the Lab. Combined provable error bounds from a previously developed LiDAR localization algorithm with Neural Network based localization algorithms. (ROS, Python)

Model Predictive Controller (MPC) for autonomous cars

UCLA, 2024

Implemented an MPC controller for waypoint tracking in an autonomous vehicle simulator. Implementation was done through dynamic linearization along the trajectory. (Python)

Simulation of Epidemics on Networks

UCLA, 2024

Simulated various testing and quarantining strategies for networked epidemiological models based on control theoretic techniques. Estimated model parameters from data using neural ODE tools. (Python)

Four Degree of Freedom Robotic Arm

UoP, 2023

Fabricated a miniature 4 DOF robotic arm from acrylic glass. Implemented pick and place motions using forward and reverse kinematics for trajectory calculation. (C++)

CAN bus based LCD for vehicle engine monitoring

UoP, 2022

Designed and assembled an LCD display to monitor engine parameters using CAN data. Used an ESP32 microcontroller for CAN decoding and LCD control. (C++)

Fetal movement monitoring using inertial sensors

UoP, 2022 & 2023

Designed and fabricated hardware and implemented algorithms for measuring abdominal vibrations in pregnant women using accelerometers, to assess fetal well-being. Developed in collaboration with Department of Obstetrics & Gynecology, Teaching Hospital Peradeniya. (Python, C++)

CCTV based crowd forecasting

UoP, 2021

Developed neural network based algorithms for predicting the the number of people present at a given location using past CCTV footage. (PyTorch, OpenCV)

Autonomous Maze Navigating Robot (Micromouse)

UoP, 2019

Designed and built custom hardware (including sensors) for an autonomous maze navigating robot. Implemented low level controllers and algorithms for navigation and motion planning. (C++)

Various Other Hobby/Course Projects

2018-2021

Analog (PD Control) Line Follower, Logic Gate level implementation of a 4-digit pass-code lock, Mini CNC machine from reused computer parts, Arcade style video game in 6502 Assembly.

SERVICES

Reviewer Roles

- Served as a reviewer for IEEE Control Systems Letters. (2025)

Volunteering

- Conducted “Introduction to python” webinars for school students and first year undergraduates hosted by the University of Peradeniya. (2021)
- Conducted Arduino and IOT workshops for school students and teachers to promote IT education among young students. (2019 & 2021)